Practical Cheat Sheet

Quick rule: Pin 3 = SDA, Pin 5 = SCL, Pin 8 = TX, Pin 10 = RX.

1. LED blinking patterns

Blinks 4 LEDs in different patterns. Uses **GPIO.BOARD** (physical pin numbers).

```
LED 1 -> Pin 7
LED 2 -> Pin 11
LED 3 -> Pin 13
LED 4 -> Pin 15
(each LED through a resistor to GND)
```

2. Telegram controls GPIO

Send "blue on" or "all off" to your bot. You need the BotFather token and the **telepot** library. Uses **GPIO.BCM**.

```
Green  +  -> Pin 3  (GPIO 2)
Yellow +  -> Pin 5  (GPIO 3)
Red    +  -> Pin 7  (GPIO 4)
Blue   +  -> Pin 11 (GPIO 17)
All negatives -> Pin 9 (GND)
```

3. TM1637 digital clock

A 4-digit display shows the current time with a blinking colon.

```
GND -> Pin 14
VCC -> Pin 4  (5V)
DIO -> Pin 18 (GPIO 24)
CLK -> Pin 16 (GPIO 23)
```

4. GPS module

Prints GPS data over serial (**/dev/serial0**, 9600 baud). TX and RX are crossed.

```
GPS GND -> Pin 6
GPS VCC -> Pin 4  (5V)
GPS RXD -> Pin 8  (TXD)
GPS TXD -> Pin 10 (RXD)
```

5. Oscilloscope (ADS1115)

The ADC reads an analog signal and draws a live graph. Enable I2C. Uses **matplotlib** and **drawnow**.

```
VDD -> Pin 17 (3.3V)
GND -> Pin 9
SCL -> Pin 5  (GPIO 3)
SDA -> Pin 3  (GPIO 2)
```

6. 16x2 I2C LCD

Shows your own text on the LCD. Check the address with **i2cdetect -y 1** (0x27 or 0x3F).

```
GND -> Pin 6
VCC -> Pin 4 (5V)
SDA -> Pin 3 (GPIO 2)
SCL -> Pin 5 (GPIO 3)
```

7. RFID read/write/erase

Reads, writes and erases tag data. Enable SPI. This module needs **3.3V, not 5V**.

```
VCC  -> Pin 1  (3.3V)
GND  -> Pin 6
SDA  -> Pin 24 (CS0)
SCK  -> Pin 23
MOSI -> Pin 19
MISO -> Pin 21
```

11. Fingerprint + LCD

Run **enroll.py** first, then **main_verify.py**. The LCD shows MATCHED or UNMATCHED. Enable both I2C and serial.

```
LCD:
GND -> Pin 6
VCC -> Pin 2 (5V)
SDA -> Pin 3
SCL -> Pin 5

Fingerprint sensor:
Red (VCC)       -> Pin 4
Black (GND)     -> Pin 9 or 14
Yellow/Green TX -> Pin 10
White/Blue RX   -> Pin 8
```

12. Fingerprint door lock (relay)

A matched fingerprint turns the relay on for 5 seconds. The relay is **active-LOW**, so LOW turns it ON.

```
Relay:
VCC -> Pin 2
GND -> Pin 9
IN  -> Pin 11 (GPIO 17)

Fingerprint sensor:
VCC -> Pin 2 (5V)
GND -> Pin 6
TX  -> Pin 10
RX  -> Pin 8
```

13. MQTT LED control

The MQTT Dashboard phone app publishes "on" or "off" to topic **led**. Broker: **broker.emqx.io**, port 1883.

```
LED (VCC) -> Pin 12 (GPIO 18)
GND       -> Pin 6
```

14. Flask web relay

A web page with ON and OFF buttons controls the relay. Open **Pi's IP:5000**.

```
VCC  -> Pin 2
GND  -> Pin 6
IN 1 -> Pin 12 (GPIO 18)
```

Your PDF's table says GPIO 18, but the code uses **RELAY = 17**. Check which pin you actually wired.

17. Node-RED button + LED

A physical button and a web dashboard both toggle the same LED. Start it with **node-red-start**.

```
Your PDF gives no pin numbers.
Push button -> any GPIO you choose
LED         -> any GPIO you choose
(set the same pins in the GPIO in / GPIO out nodes)
```

18. CPU%, memory%, temperature

Prints CPU usage, memory usage and temperature every second. Uses **psutil**.

```
No connections (software only)
```

19. Access control system

RFID card or fingerprint unlocks the relay. The LCD shows the status and every attempt is saved to a CSV log. Run programme 01 first to get your card ID for **WHITELIST**.

```
Your PDF gives no connection table.
Code pins: Relay -> GPIO 23, LED -> GPIO 21
LCD, RFID and fingerprint sensor: wired as in practicals 11 and 7
```

Top 3 exam mistakes: mixing up BOARD and BCM numbering, forgetting to enable I2C/SPI/serial, and giving the RFID module 5V.